

# Bloom Filter-Based MPSI

Weekly Progress Meeting 2 April 2026

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Daily Supervisor: Tjitske Koster

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# Bibliography

- [1] Jelle Vos, Jorrit van Assen, Tjitske Koster, Evangelia Anna Markatou, and Zekeriya Erkin. On the insecurity of bloom filter-based private set intersections. *Cryptology ePrint Archive*, 2024.
- [2] Ou Ruan, Changwang Yan, Jing Zhou, and Chaohao Ai. A practical multiparty private set intersection protocol based on bloom filters for unbalanced scenarios. *Applied Sciences*, 13(24):13215, 2023.
- [3] Florian Kerschbaum. Outsourced private set intersection using homomorphic encryption. In *Proceedings of the 7<sup>th</sup> ACM Symposium on Information, Computer and Communications Security*, pages 85–86, 2012.
- [4] Falzon, Francesca, and Evangelia Anna Markatou. "Re-visiting authorized private set intersection: A new privacy-preserving variant and two protocols." *Proceedings on Privacy Enhancing Technologies* (2025).
- [5] Koster, Tjitske Ollie, Francesca Falzon, and Evangelia Anna Markatou. "Bandwidth Efficient Partial Authorized PSI." *Cryptology ePrint Archive* (2025).
- [6] Lindell, Yehuda. "How to simulate it—a tutorial on the simulation proof technique." *Tutorials on the Foundations of Cryptography: Dedicated to Oded Goldreich* (2017): 277-346.

# Progress

- Implement PAPS12
  - Look more closely and try to understand the code of [4] (Lilika's APSI & PAPS1) and [5] (Tjitske's PAPS1)
- Brainstorm on APSI1
- Start thinking about the proof for APSI2
  - Finish reading [6] ("How to simulate it—a tutorial on the simulation proof technique")

# Progress

## APSI 2 – Interactive Judge

$n_t$  = server's set size

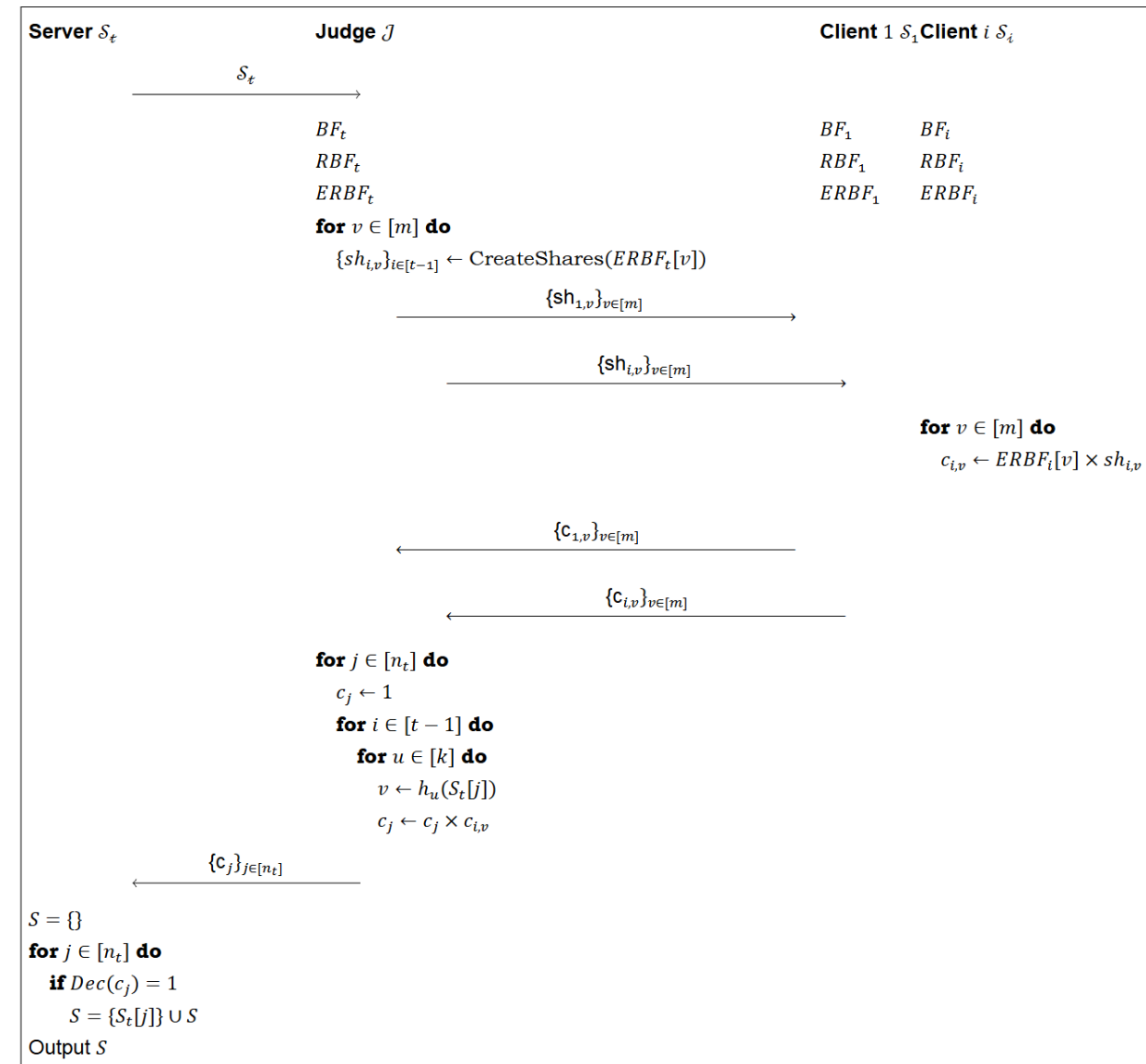
$m$  = bin count (BF size)

$k$  = number of hash functions

$t$  = number of parties (including the server)

$ERBF_i$  = the encrypted randomized Bloom filter of party  $i$

Only the Server has the decryption key

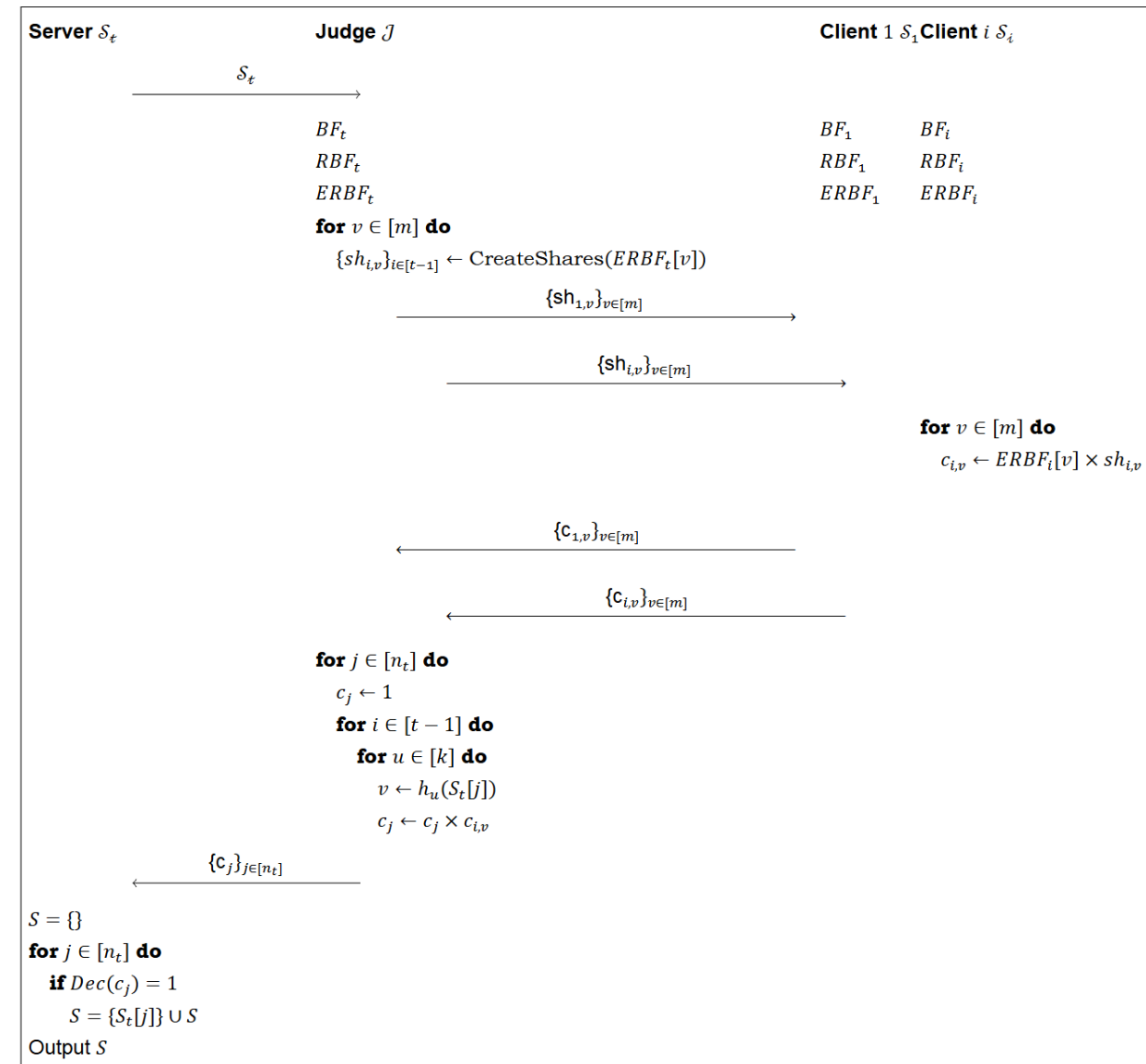


# Progress

## APSI 2 – Interactive Judge

Threat Model:

- Malicious Server
- (Semi?-)Honest Judge
- Semi-honest clients



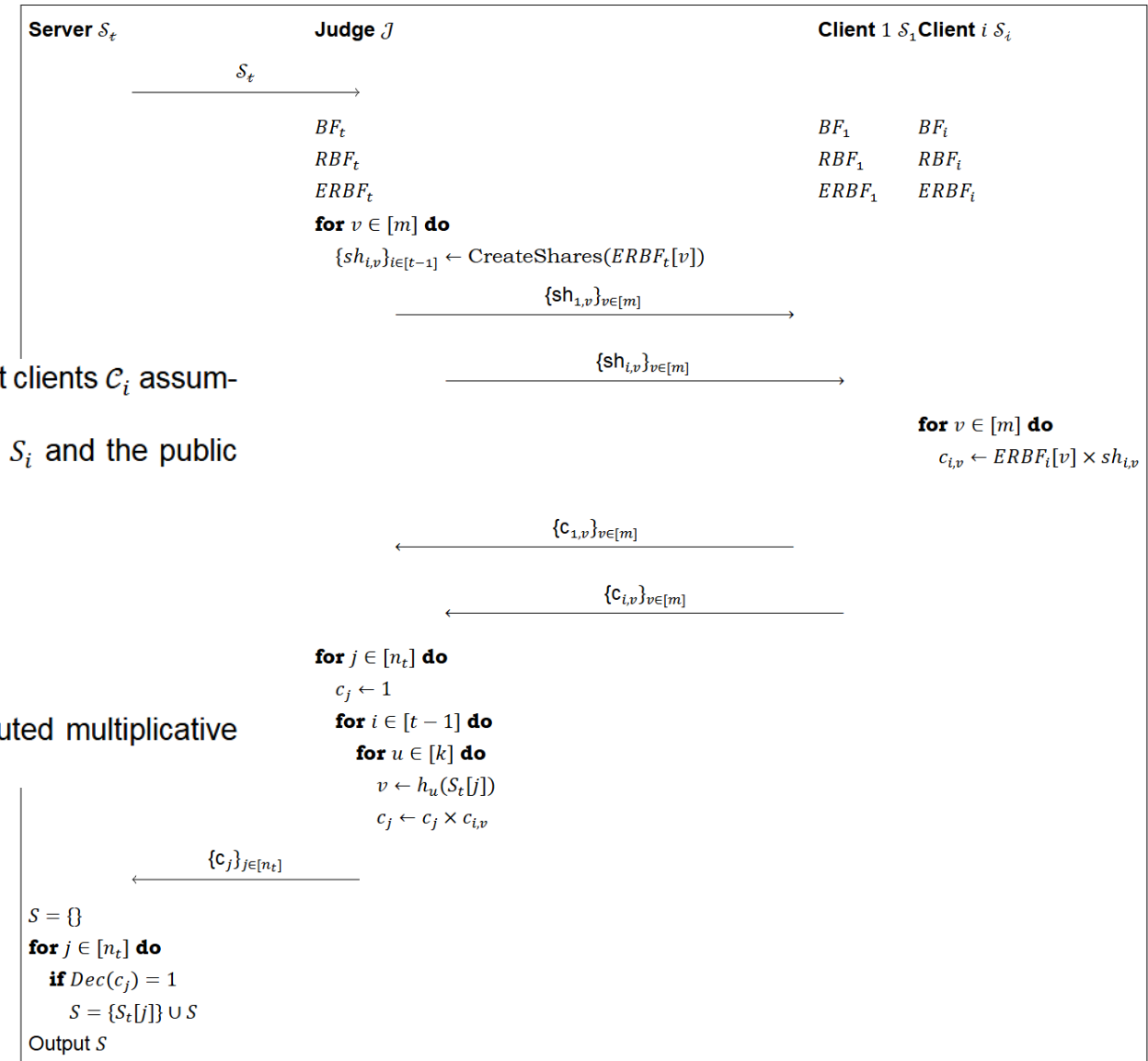
# Progress

## APSI 2 – Interactive Judge

**Security against Semi-Honest Client  $\mathcal{C}_i$**  The protocol is secure against semi-honest clients  $\mathcal{C}_i$  assuming the hiding property of the underlying secret sharing scheme.

**Proof:** We construct a simulator  $Sim_{\mathcal{C}_i}$  that takes as input the client's own set  $S_i$  and the public parameters. The Simulator ( $Sim_{\mathcal{C}_i}$ ) proceeds as follows:

1. Define  $BF_t$  as empty by filling in each bin  $v \in [m]$  with 0.
2. Define  $RBF_t$  by sampling random values for each bin  $v \in [m]$ .
3. Define  $ERBF_t$  by encrypting each bin  $v \in [m]$  of  $RBF_t$ .
4. For each bin  $ERBF_i[v]$  with  $v \in [m]$ ,  $Sim_{\mathcal{C}_i}$  generate  $t - 1$  uniformly distributed multiplicative shares  $\{sh_{i,v}\}_{i \in [t]}$  and send share  $sh_{i,v}$  to  $\mathcal{C}_i$ .

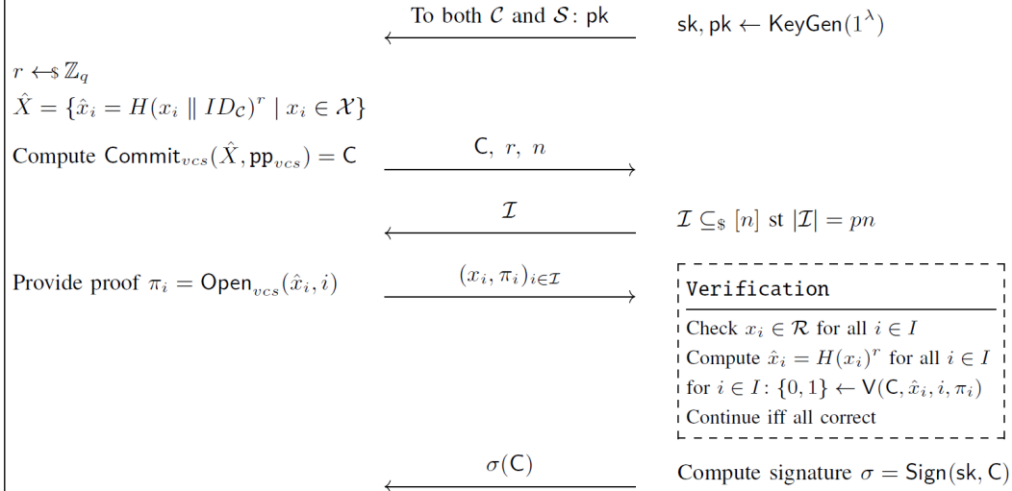


# PAPSI 2

## Authorize

Receiver  $\mathcal{C}(\mathcal{X} = \{x_i\}_{i \in [n]})$

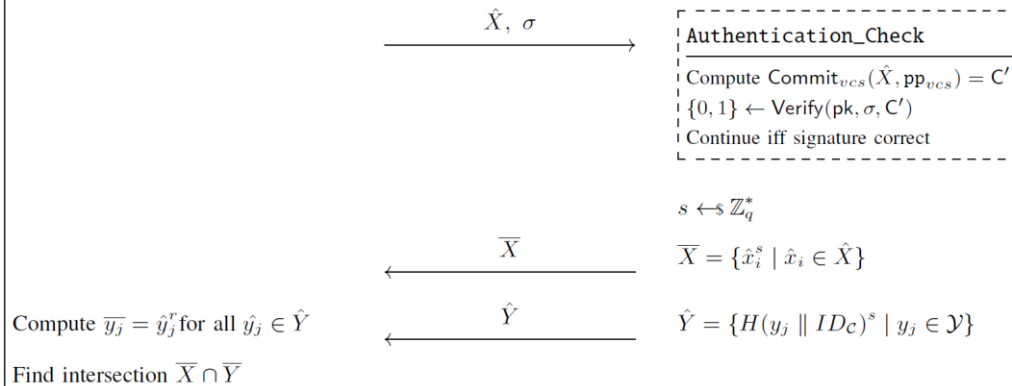
Judge



## Intersect

Receiver

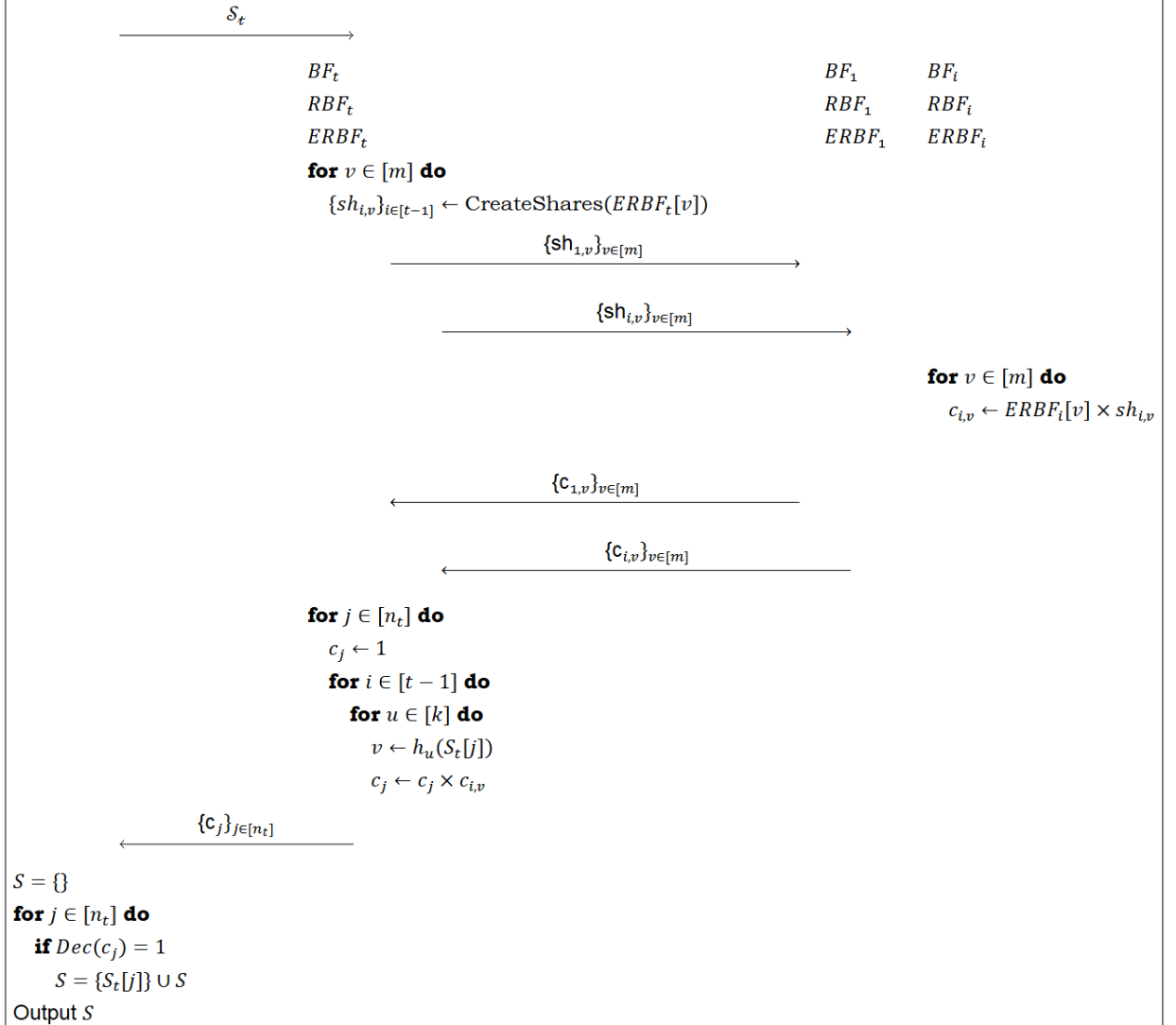
Sender  $\mathcal{S}(\mathcal{Y} = \{y_i\}_{i \in [m]})$



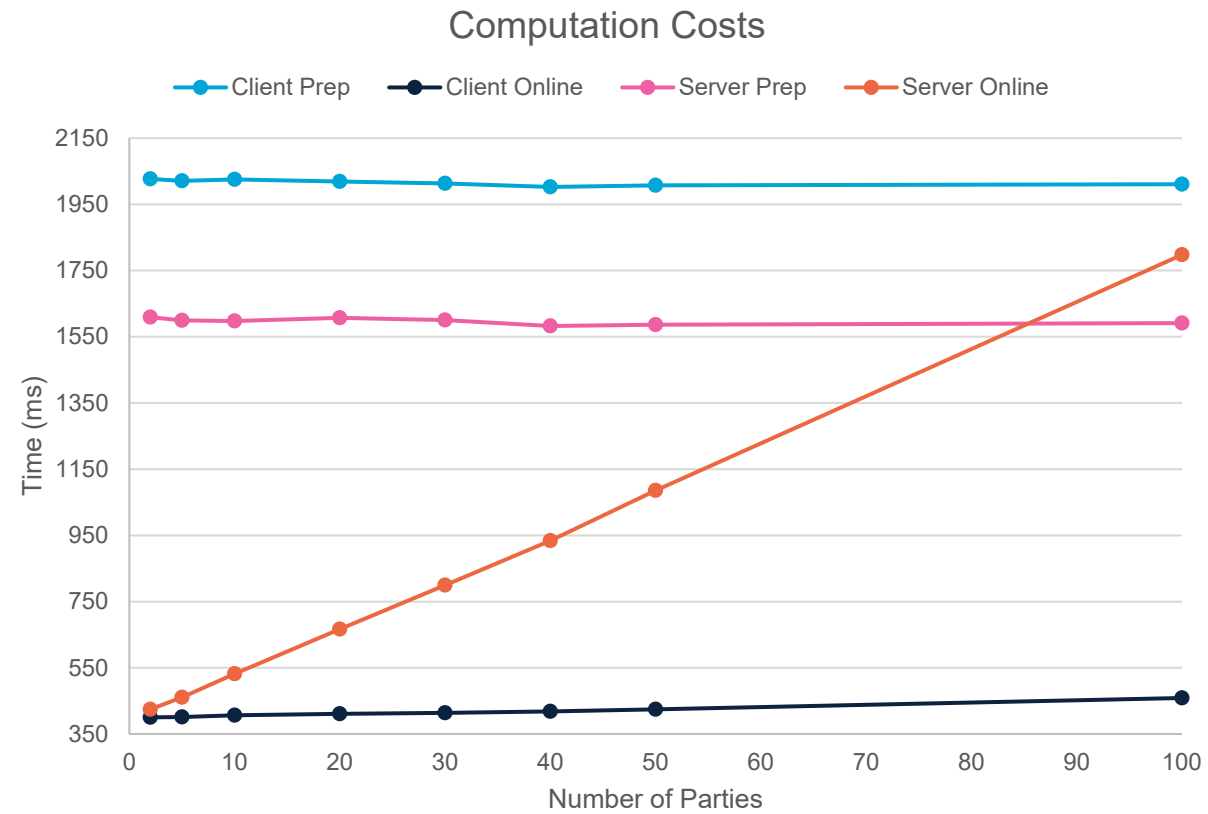
Server  $\mathcal{S}_t$

Judge  $\mathcal{J}$

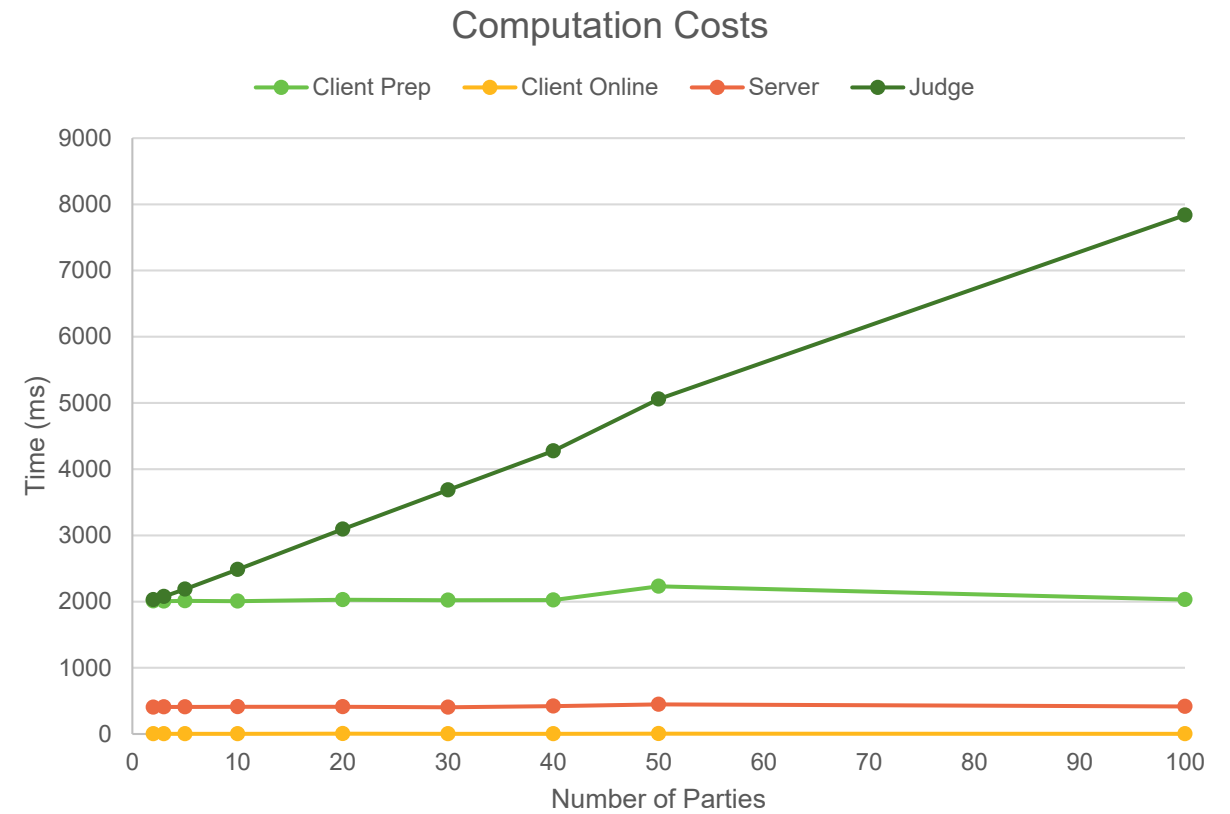
Client 1  $\mathcal{S}_1$  Client  $i$   $\mathcal{S}_i$



# Baseline

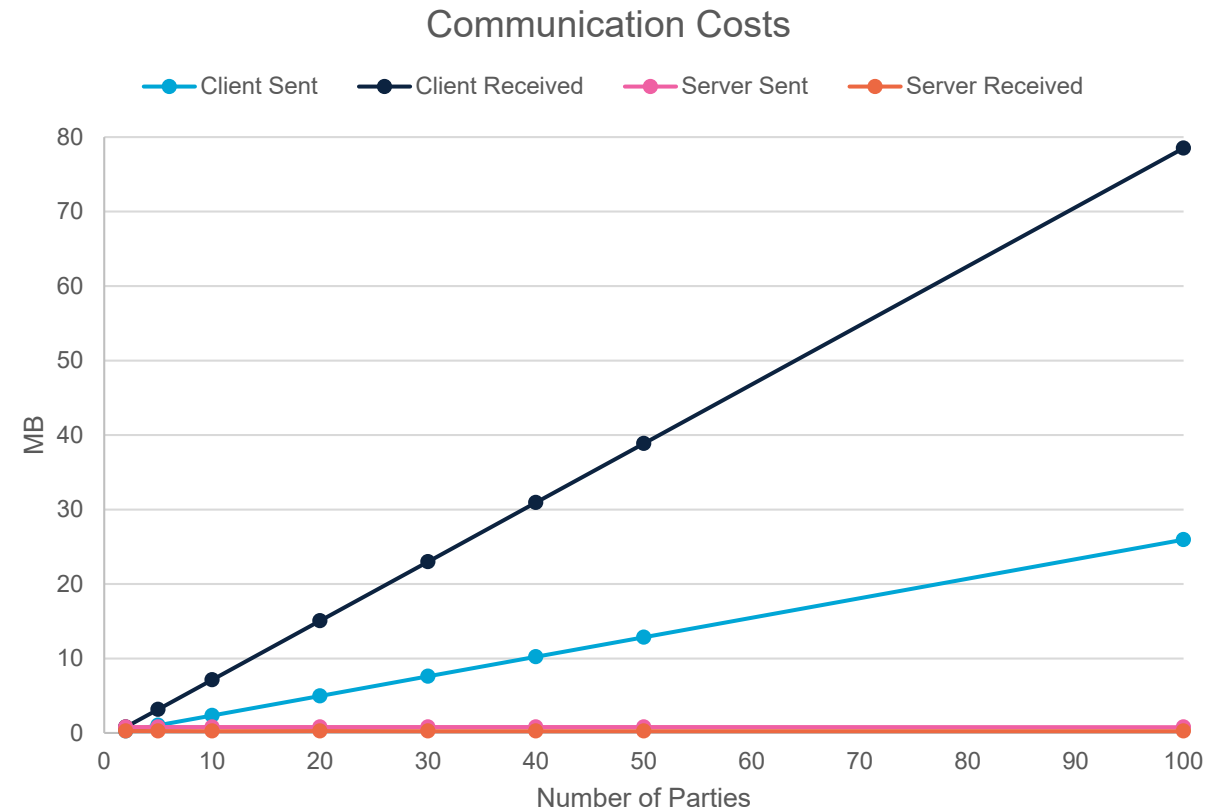


# APSI2

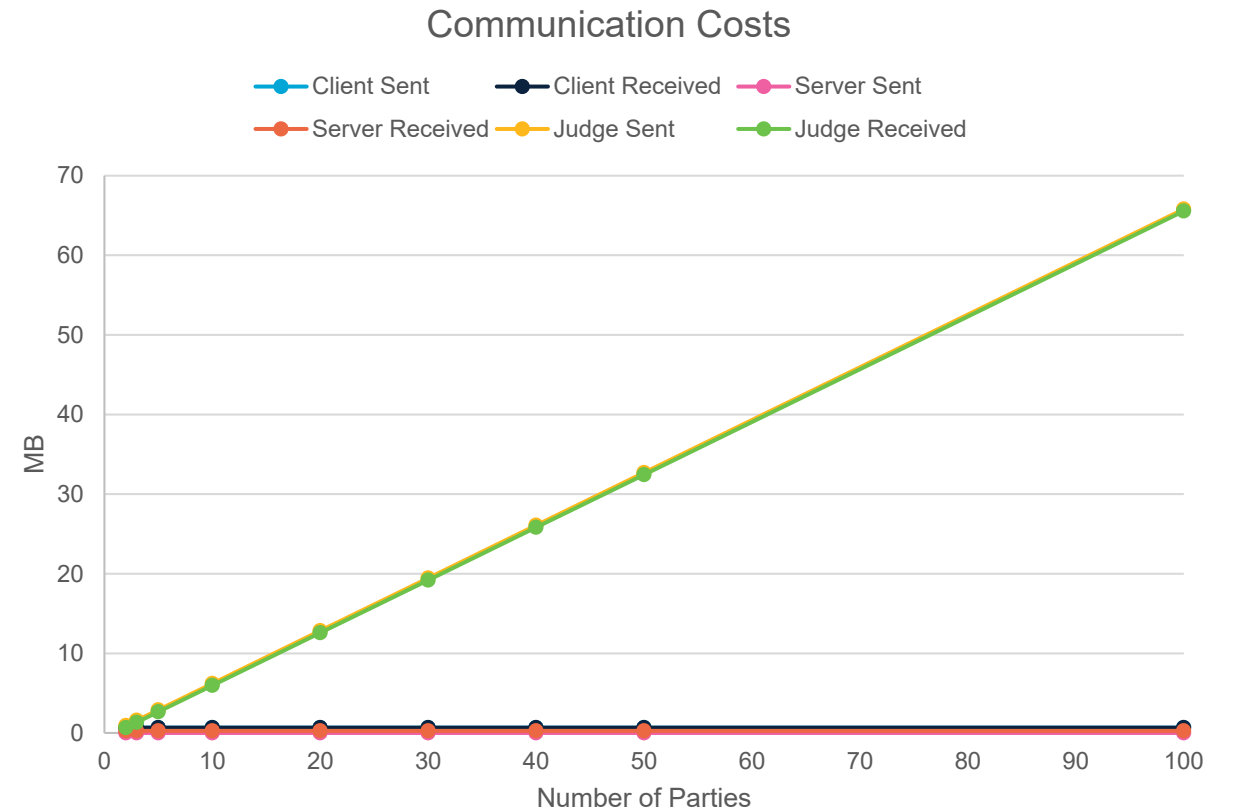




# Baseline

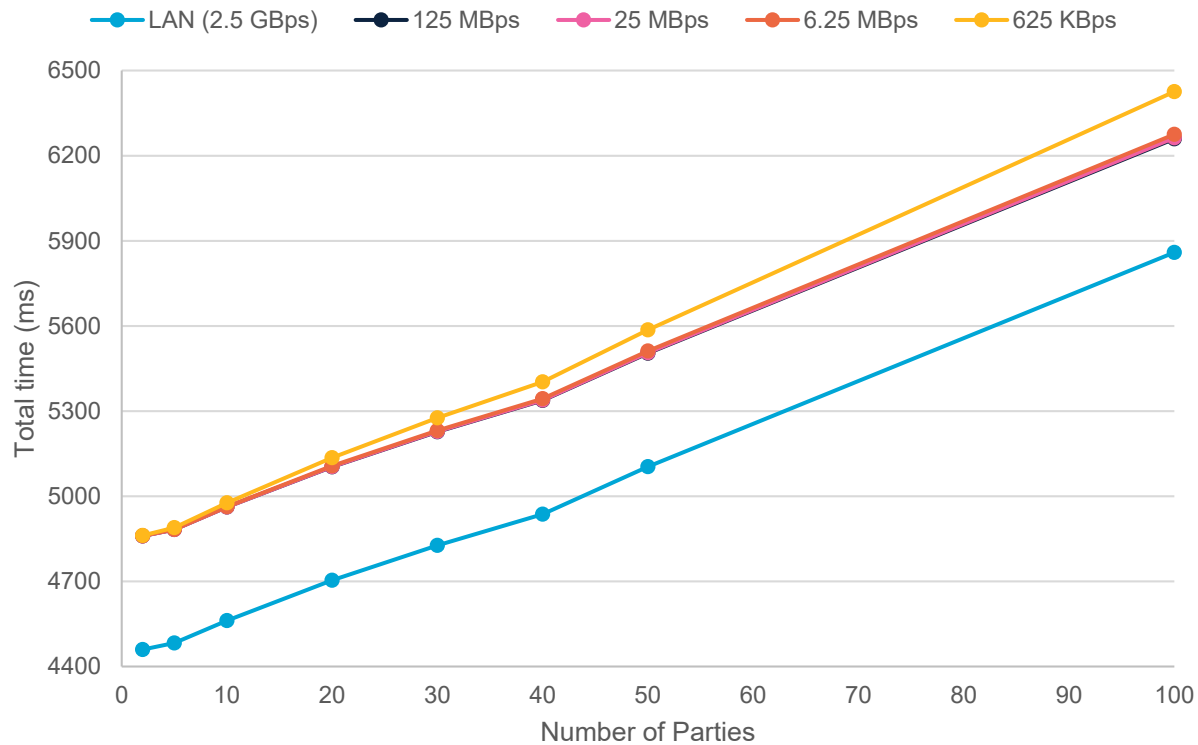


# APSI2



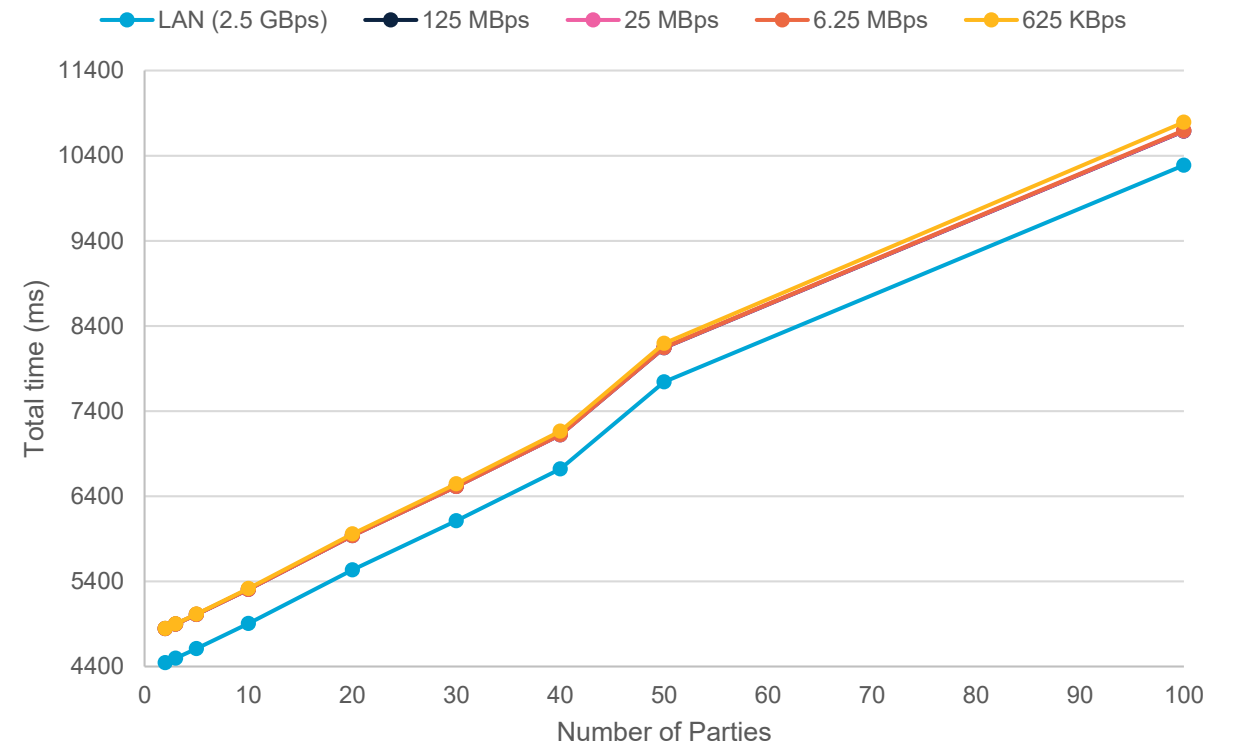
# Baseline

Simulated Total Execution Time



# APSI2

Simulated Total Execution Time



# Next Week

- Implement PAPS12
- Brainstorm on APSI1
- Continue working on the proof for APSI2

# Thank you!

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